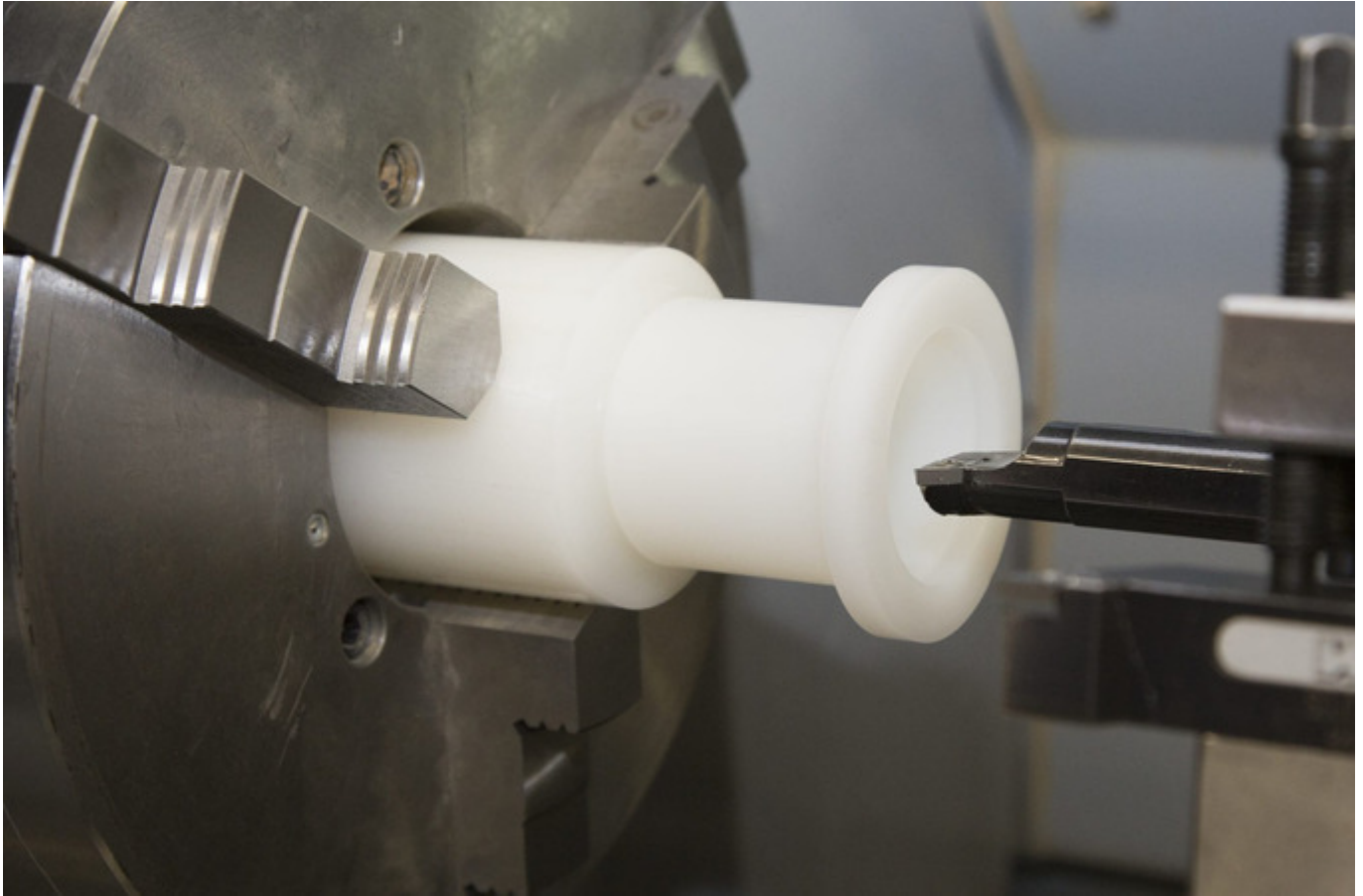


Qualification Pack



Machine Operator - CNC Turning of Plastic (Lathe)

QP Code: RSC/Q9901

Version: 1.0

NSQF Level: 4

Rubber, Chemical & Petrochemical Skill development Council || 217, 2nd floor, Rectangle One, Saket
District Centre
New Delhi - 110017 || email:vishaka.v@rcpsdc.in

Qualification Pack

Contents

RSC/Q9901: Machine Operator - CNC Turning of Plastic (Lathe)	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
RSC/N9904: Interpret CNC Programs and drawings and set up CNC Turning Machine	5
RSC/N9905: Perform turning operations on Plastic Workpieces using CNC Machine	12
RSC/N9906: Maintain CNC Turning machine, tools and work area	19
RSC/N5007: Follow health & safety protocol	26
DGT/VSQ/N0101: Employability Skills (30 Hours)	34
Assessment Guidelines and Weightage	39
<i>Assessment Guidelines</i>	39
<i>Assessment Weightage</i>	40
Acronyms	41
Glossary	42

Qualification Pack

RSC/Q9901: Machine Operator - CNC Turning of Plastic (Lathe)

Brief Job Description

A Machine Operator CNC Turning of Plastic sets up and operates CNC turning machines to produce precision plastic components as per engineering drawings and specifications. The role involves interpreting CNC programs, selecting tools, preparing raw materials, conducting machining operations, inspecting output, and maintaining machines, tools, and work areas. The operator ensures safety, quality, and productivity in line with organizational and NSQF standards.

Personal Attributes

The individual should have good hand eye coordination, attention to detail, and problem-solving ability to perform precise machining operations. They must demonstrate discipline, responsibility, and commitment to safety while handling machines and plastic materials. Basic numerical ability, spatial visualization, and the capacity to follow instructions and work as part of a team are essential. Physical stamina, patience, and willingness to learn new machining practices are also important attributes.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [RSC/N9904: Interpret CNC Programs and drawings and set up CNC Turning Machine](#)
2. [RSC/N9905: Perform turning operations on Plastic Workpieces using CNC Machine](#)
3. [RSC/N9906: Maintain CNC Turning machine, tools and work area](#)
4. [RSC/N5007: Follow health & safety protocol](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Rubber
Sub-Sector	Tool and Die Moulding
Occupation	CNC Machining and Die Manufacturing
Country	India
NSQF Level	4

Qualification Pack

Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7223.6002
Minimum Educational Qualification & Experience	12th grade Pass (or Equivalent) OR 10th grade pass with 3 Years of experience relevant OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience relevant OR 10th grade pass plus 2-year NTC (/NAC/CITS or Equivalent)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	30/04/2025
Version	1.0
Reference code on NQR	QG-04-RI-04050-2025-V2-RSDC
NQR Version	2

Qualification Pack

RSC/N9904: Interpret CNC Programs and drawings and set up CNC Turning Machine

Description

This NOS unit covers interpreting CNC programs and engineering drawings and setting up CNC turning machines for plastic components. It includes reading drawings, verifying codes, setting tools and work offsets, conducting trial runs, and ensuring safety, quality, and machine readiness.

Scope

The scope covers the following :

- Interpret CNC Programs and Drawings
- Set Up CNC Turning Machine to the Job

Elements and Performance Criteria

Interpret CNC Programs and Drawings

To be competent, the user/individual on the job must be able to:

- PC1.** read and interpret engineering drawings, job sheets, and process documents for turning plastics
- PC2.** identify dimensions, tolerances, fits, and surface finish symbols accurately
- PC3.** interpret sectional views, auxiliary views, and geometric dimensioning details relevant to machining
- PC4.** extract relevant information from 2D sketches and 3D CAD models of the component
- PC5.** recognize machining allowances, reference points, and datum surfaces from drawings/models
- PC6.** read ISO codes, G-codes, and M-codes used in CNC turning programs
- PC7.** understand tool path sequence, cycle codes, and program structure specific to plastics machining
- PC8.** cross-check program codes against drawing dimensions and process sheet requirements
- PC9.** ensure correct tool number, offsets, spindle speed, and feed values are used for plastic components
- PC10.** consider material properties of plastics (softening point, chip formation, thermal expansion) during machining planning
- PC11.** identify suitable cutting parameters to minimize deformation, melting, or surface damage
- PC12.** identify errors in CNC program (e.g., missing codes, incorrect tool paths, speed/feed mismatch)
- PC13.** detect inconsistencies between drawing, program, and process sheet and escalate/report to supervisor
- PC14.** carry out minor program edits within defined scope (e.g., tool offset correction, feed/speed adjustment)
- PC15.** verify revised code through dry run or simulation before machining

Qualification Pack

Set Up CNC Turning Machine to the Job

To be competent, the user/individual on the job must be able to:

- PC16.** check the CNC turning machine for cleanliness, lubrication, and proper functioning before setup
- PC17.** ensure safety guards, interlocks, and emergency stop functions are in working order
- PC18.** identify suitable cutting tools and tool holders for plastic machining as per job requirement
- PC19.** mount and secure tools and holders correctly, ensuring proper nose orientation and rigidity
- PC20.** select and set up appropriate work-holding devices (chuck, collet, fixture) suitable for plastic workpieces
- PC21.** verify alignment, concentricity, and secure clamping to avoid deformation of plastic material
- PC22.** load raw plastic rods, billets, or bars into the work-holding device following standard procedure
- PC23.** check correct positioning, alignment, and safe clamping without damaging the material
- PC24.** transfer or input CNC program into the control system accurately
- PC25.** verify the program sequence, tool call-up, spindle speed, and feed rates against the job sheet
- PC26.** establish machine zero, work offsets, and tool offsets using appropriate measuring instruments
- PC27.** recheck offsets and update in the system to ensure dimensional accuracy
- PC28.** set spindle speed, feed rate, depth of cut, and coolant/lubrication parameters appropriate for plastics
- PC29.** ensure parameters minimize tool wear, melting, or surface defects
- PC30.** carry out a dry run or trial cut to confirm accuracy of setup and program
- PC31.** inspect the trial component using gauges, micrometers, or calipers to verify dimensions
- PC32.** make minor adjustments in offsets, parameters, or clamping if required
- PC33.** clean chips, burs, and debris from machine area after setup
- PC34.** carry out preventive checks and ensure machine is in ready-to-operate condition for production

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** how to read and interpret engineering drawings, symbols, tolerances, GD&T, and surface finish requirements
- KU2.** how to extract key details from 2D/3D CAD models and job sheets for machining plastics
- KU3.** the structure and functions of CNC programs, including ISO standards, G-codes, M-codes, and cycle codes
- KU4.** how to verify CNC program steps against drawings, process sheets, and tooling layouts
- KU5.** properties of plastics (softening, melting point, chip formation, thermal expansion) and their effect on machining parameters
- KU6.** selection criteria for cutting tools, tool holders, inserts, and work-holding devices suitable for plastics

Qualification Pack

- KU7.** procedures for mounting, aligning, and clamping tools and workpieces to avoid deformation or damage
- KU8.** methods to set reference points, tool offsets, and work offsets on CNC turning machines
- KU9.** how to select and adjust spindle speed, feed rates, and depth of cut for plastic machining
- KU10.** importance and use of coolants, lubricants, or dry machining for different plastic materials
- KU11.** techniques for conducting dry runs and trial machining to validate setup and programs
- KU12.** inspection methods using micrometers, calipers, gauges, and other measuring instruments to verify accuracy
- KU13.** machine maintenance practices including cleaning, lubrication, and preventive checks
- KU14.** safety practices, use of PPE, machine guarding, and emergency stop procedures while operating CNC machines
- KU15.** documentation and reporting procedures for program edits, inspection results, and production records

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate clearly with supervisors, peers, and quality inspectors regarding job instructions and program issues
- GS2.** interpret technical terms, symbols, and instructions in job sheets, SOPs, and process documents
- GS3.** apply basic mathematical calculations (fractions, decimals, geometry, trigonometry) for dimensions and tool offsets
- GS4.** use measuring instruments such as vernier calipers, micrometers, dial gauges, and CMM effectively
- GS5.** operate CNC machine control panels and input/edit CNC codes with accuracy
- GS6.** conduct logical troubleshooting of machining errors (tool wear, dimensional deviation, surface defects)
- GS7.** adapt machining parameters based on different grades and behaviors of plastics
- GS8.** perform safe handling, clamping, and loading of plastic workpieces without causing damage
- GS9.** demonstrate problem-solving skills to identify root causes and implement corrective actions
- GS10.** work independently as well as in a team to meet production schedules and quality requirements
- GS11.** maintain a high level of attention to detail for quality, safety, and process adherence
- GS12.** follow workplace safety norms, use PPE, and comply with environmental practices in plastic machining
- GS13.** document program changes, inspection results, and production data as per organizational procedures
- GS14.** plan and organize work to utilize machine time efficiently and minimize downtime
- GS15.** apply continuous learning skills to keep updated with new CNC technologies, plastic machining methods, and quality standards

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Interpret CNC Programs and Drawings</i>	18	30	-	-
PC1. read and interpret engineering drawings, job sheets, and process documents for turning plastics	1	2	-	-
PC2. identify dimensions, tolerances, fits, and surface finish symbols accurately	1	2	-	-
PC3. interpret sectional views, auxiliary views, and geometric dimensioning details relevant to machining	2	2	-	-
PC4. extract relevant information from 2D sketches and 3D CAD models of the component	1	2	-	-
PC5. recognize machining allowances, reference points, and datum surfaces from drawings/models	1	2	-	-
PC6. read ISO codes, G-codes, and M-codes used in CNC turning programs	1	2	-	-
PC7. understand tool path sequence, cycle codes, and program structure specific to plastics machining	2	2	-	-
PC8. cross-check program codes against drawing dimensions and process sheet requirements	1	2	-	-
PC9. ensure correct tool number, offsets, spindle speed, and feed values are used for plastic components	1	2	-	-
PC10. consider material properties of plastics (softening point, chip formation, thermal expansion) during machining planning	1	2	-	-
PC11. identify suitable cutting parameters to minimize deformation, melting, or surface damage	2	2	-	-
PC12. identify errors in CNC program (e.g., missing codes, incorrect tool paths, speed/feed mismatch)	1	2	-	-
PC13. detect inconsistencies between drawing, program, and process sheet and escalate/report to supervisor	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. carry out minor program edits within defined scope (e.g., tool offset correction, feed/speed adjustment)	1	2	-	-
PC15. verify revised code through dry run or simulation before machining	1	2	-	-
<i>Set Up CNC Turning Machine to the Job</i>	22	30	-	-
PC16. check the CNC turning machine for cleanliness, lubrication, and proper functioning before setup	1	2	-	-
PC17. ensure safety guards, interlocks, and emergency stop functions are in working order	2	2	-	-
PC18. identify suitable cutting tools and tool holders for plastic machining as per job requirement	1	1	-	-
PC19. mount and secure tools and holders correctly, ensuring proper nose orientation and rigidity	1	2	-	-
PC20. select and set up appropriate work-holding devices (chuck, collet, fixture) suitable for plastic workpieces	1	1	-	-
PC21. verify alignment, concentricity, and secure clamping to avoid deformation of plastic material	1	2	-	-
PC22. load raw plastic rods, billets, or bars into the work-holding device following standard procedure	1	1	-	-
PC23. check correct positioning, alignment, and safe clamping without damaging the material	2	2	-	-
PC24. transfer or input CNC program into the control system accurately	1	1	-	-
PC25. verify the program sequence, tool call-up, spindle speed, and feed rates against the job sheet	1	2	-	-
PC26. establish machine zero, work offsets, and tool offsets using appropriate measuring instruments	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. recheck offsets and update in the system to ensure dimensional accuracy	1	2	-	-
PC28. set spindle speed, feed rate, depth of cut, and coolant/lubrication parameters appropriate for plastics	1	1	-	-
PC29. ensure parameters minimize tool wear, melting, or surface defects	1	2	-	-
PC30. carry out a dry run or trial cut to confirm accuracy of setup and program	1	1	-	-
PC31. inspect the trial component using gauges, micrometers, or calipers to verify dimensions	2	2	-	-
PC32. make minor adjustments in offsets, parameters, or clamping if required	1	2	-	-
PC33. clean chips, burs, and debris from machine area after setup	1	1	-	-
PC34. carry out preventive checks and ensure machine is in ready-to-operate condition for production	1	2	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N9904
NOS Name	Interpret CNC Programs and drawings and set up CNC Turning Machine
Sector	Rubber
Sub-Sector	
Occupation	CNC Machining and Die Manufacturing
NSQF Level	4
Credits	5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

RSC/N9905: Perform turning operations on Plastic Workpieces using CNC Machine

Description

This NOS unit covers performing CNC turning operations on plastic workpieces, including facing, turning, grooving, threading, drilling, boring, and parting-off while applying correct machining parameters. It emphasizes quality, safety, and monitoring to achieve accurate dimensions and surface finish without damaging the plastic material.

Scope

The scope covers the following :

- Carry out CNC Turning Operations on Plastic Workpieces
- Inspect and Ensure Quality of Turned Plastic Components

Elements and Performance Criteria

Carry out CNC Turning Operations on Plastic Workpieces

To be competent, the user/individual on the job must be able to:

- PC1.** carry out facing to achieve smooth, flat surfaces on plastic workpieces
- PC2.** perform straight turning and taper turning as per specified dimensions
- PC3.** cut external and internal grooves accurately to drawing specifications
- PC4.** perform threading, boring, drilling, and parting-off operations on plastic rods and billets
- PC5.** select and apply spindle speed, feed rate, and depth of cut appropriate for plastic machining
- PC6.** adjust machining parameters to prevent melting, deformation, or surface burns on plastics
- PC7.** continuously observe chip formation, surface finish, and dimensional accuracy during operations
- PC8.** monitor tool wear, vibration, and machine sounds to identify abnormal conditions
- PC9.** apply dry machining, compressed air, or coolant/lubricant as per plastic material requirements
- PC10.** ensure coolant/lubricant does not chemically affect the plastic workpiece
- PC11.** load, clamp, and handle plastic workpieces carefully to avoid scratches, dents, or distortion
- PC12.** follow safety protocols while machining, including PPE use and machine guarding
- PC13.** achieve required tolerances, dimensions, and surface quality as per job sheet and drawing
- PC14.** take corrective actions (offset adjustments, tool replacement) when deviations are observed
- PC15.** adhere to SOPs, process sheets, and quality norms while carrying out CNC turning operations
- PC16.** Maintain documentation of operations performed, adjustments made, and output produced

Inspect and Ensure Quality of Turned Plastic Components

To be competent, the user/individual on the job must be able to:

- PC17.** measure dimensions of turned plastic components using calipers, micrometers, height gauges, and bore gauges

Qualification Pack

- PC18.** check geometrical tolerances such as concentricity, cylindricity, straightness, and run-out
- PC19.** assess surface finish of machined plastic parts using appropriate instruments or visual standards
- PC20.** verify measurements against engineering drawings, CAD models, and job sheets
- PC21.** ensure that tolerances, fits, and surface finish requirements are achieved as per quality standards
- PC22.** detect machining errors such as burrs, scratches, chatter marks, or dimensional deviations
- PC23.** analyze possible causes of deviations (tool wear, incorrect offsets, improper clamping, or wrong parameters)
- PC24.** make fine adjustments to tool offsets, machining parameters, or process sequence to correct deviations
- PC25.** replace worn-out cutting tools or inserts when required
- PC26.** report persistent issues or non-conformities to the supervisor/quality inspector
- PC27.** record measurement readings, deviations, and corrective actions in inspection logs
- PC28.** maintain production quality records as per organizational formats and standards
- PC29.** confirm that machined plastic components meet organizational and customer quality standards
- PC30.** approve only conforming parts for downstream processes or customer delivery
- PC31.** segregate and label non-conforming parts for rework or rejection as per company procedure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** how to read and interpret engineering drawings, CAD models, and job sheets for plastic components
- KU2.** how to identify dimensions, tolerances, and surface finish symbols relevant to CNC turning
- KU3.** the properties of plastics such as thermal expansion, softening temperature, and machinability
- KU4.** the sequence and application of CNC turning operations like facing, taper turning, grooving, threading, boring, and parting-off
- KU5.** how to apply suitable spindle speed, feed rate, and depth of cut for different plastic materials
- KU6.** the selection and use of cutting tools, inserts, and tool holders appropriate for machining plastics
- KU7.** how to set up and secure plastic rods, billets, and tubes in work-holding devices without causing deformation
- KU8.** the correct use of dry machining, compressed air, or coolant/lubricant as per plastic material requirements
- KU9.** how to monitor chip formation, tool wear, and machine performance during operations
- KU10.** the common machining defects in plastics such as burrs, melting, chatter marks, and dimensional inaccuracies
- KU11.** how to operate CNC machine controls, including program execution, cycle start/stop, and emergency stop functions

Qualification Pack

- KU12.** the importance of safety practices, PPE usage, and hazard prevention while operating CNC machines
- KU13.** how to carry out routine machine checks, lubrication, and cleaning for operational readiness
- KU14.** the organizations standard operating procedures (SOPs), quality norms, and documentation requirements
- KU15.** how to report deviations, equipment malfunctions, or non-conformities to supervisors

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** apply basic mathematical calculations for dimensions, feeds, speeds, and tolerances
- GS2.** use measurement and inspection tools such as calipers, micrometers, gauges, and surface finish testers
- GS3.** communicate clearly with supervisors, quality inspectors, and team members regarding job requirements or deviations
- GS4.** interpret technical instructions, job sheets, process sheets, and tool layouts
- GS5.** identify and resolve routine issues such as tool wear, incorrect offsets, or minor machine alarms
- GS6.** plan and organize turning operations to achieve productivity and quality targets
- GS7.** follow workplace safety guidelines, use PPE, and respond effectively to emergency situations
- GS8.** handle plastic raw materials and finished components carefully to prevent damage or contamination
- GS9.** maintain accurate records of machining parameters, inspection results, and production output
- GS10.** use basic computer functions and CNC control panel interfaces for program input and operation
- GS11.** adapt machining parameters based on plastic material behavior and part geometry
- GS12.** work independently on assigned operations while coordinating effectively with team members
- GS13.** identify opportunities for improving machining efficiency, tool life, and quality
- GS14.** maintain discipline, punctuality, and responsibility as per organizational standards
- GS15.** demonstrate problem-solving skills in troubleshooting machining issues and ensuring conformance to quality

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out CNC Turning Operations on Plastic Workpieces</i>	20	30	-	-
PC1. carry out facing to achieve smooth, flat surfaces on plastic workpieces	1	2	-	-
PC2. perform straight turning and taper turning as per specified dimensions	2	2	-	-
PC3. cut external and internal grooves accurately to drawing specifications	1	1	-	-
PC4. perform threading, boring, drilling, and parting-off operations on plastic rods and billets	1	2	-	-
PC5. select and apply spindle speed, feed rate, and depth of cut appropriate for plastic machining	1	2	-	-
PC6. adjust machining parameters to prevent melting, deformation, or surface burns on plastics	2	1	-	-
PC7. continuously observe chip formation, surface finish, and dimensional accuracy during operations	1	2	-	-
PC8. monitor tool wear, vibration, and machine sounds to identify abnormal conditions	1	2	-	-
PC9. apply dry machining, compressed air, or coolant/lubricant as per plastic material requirements	1	2	-	-
PC10. ensure coolant/lubricant does not chemically affect the plastic workpiece	1	2	-	-
PC11. load, clamp, and handle plastic workpieces carefully to avoid scratches, dents, or distortion	1	2	-	-
PC12. follow safety protocols while machining, including PPE use and machine guarding	2	2	-	-
PC13. achieve required tolerances, dimensions, and surface quality as per job sheet and drawing	1	2	-	-
PC14. take corrective actions (offset adjustments, tool replacement) when deviations are observed	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. adhere to SOPs, process sheets, and quality norms while carrying out CNC turning operations	2	2	-	-
PC16. Maintain documentation of operations performed, adjustments made, and output produced	1	2	-	-
<i>Inspect and Ensure Quality of Turned Plastic Components</i>	20	30	-	-
PC17. measure dimensions of turned plastic components using calipers, micrometers, height gauges, and bore gauges	1	2	-	-
PC18. check geometrical tolerances such as concentricity, cylindricity, straightness, and run-out	1	2	-	-
PC19. assess surface finish of machined plastic parts using appropriate instruments or visual standards	1	2	-	-
PC20. verify measurements against engineering drawings, CAD models, and job sheets	2	2	-	-
PC21. ensure that tolerances, fits, and surface finish requirements are achieved as per quality standards	1	2	-	-
PC22. detect machining errors such as burrs, scratches, chatter marks, or dimensional deviations	1	2	-	-
PC23. analyze possible causes of deviations (tool wear, incorrect offsets, improper clamping, or wrong parameters)	1	2	-	-
PC24. make fine adjustments to tool offsets, machining parameters, or process sequence to correct deviations	2	2	-	-
PC25. replace worn-out cutting tools or inserts when required	1	2	-	-
PC26. report persistent issues or non-conformities to the supervisor/quality inspector	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. record measurement readings, deviations, and corrective actions in inspection logs	2	2	-	-
PC28. maintain production quality records as per organizational formats and standards	1	2	-	-
PC29. confirm that machined plastic components meet organizational and customer quality standards	2	2	-	-
PC30. approve only conforming parts for downstream processes or customer delivery	1	2	-	-
PC31. segregate and label non-conforming parts for rework or rejection as per company procedure	2	2	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N9905
NOS Name	Perform turning operations on Plastic Workpieces using CNC Machine
Sector	Rubber
Sub-Sector	
Occupation	CNC Machining and Die Manufacturing
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

RSC/N9906: Maintain CNC Turning machine, tools and work area

Description

This NOS unit covers maintaining CNC turning machines, cutting tools, and the work area to ensure safe, efficient, and uninterrupted operations. It includes cleaning, lubrication, preventive checks, tool care, and adherence to workplace safety and housekeeping standards.

Scope

The scope covers the following :

- Maintain CNC Turning Machine and Tools
- Maintain Work Area and Safety Compliance

Elements and Performance Criteria

Maintain CNC Turning Machine and Tools

To be competent, the user/individual on the job must be able to:

- PC1.** remove plastic chips, dust, and debris from machine surfaces and enclosures
- PC2.** clean tool posts, work-holding fixtures, and guards to maintain accuracy and safety
- PC3.** dispose of plastic waste and scrap in designated bins following organizational procedure
- PC4.** check and top up lubrication oil, coolant, or compressed air systems as specified by the manufacturer
- PC5.** Perform routine preventive checks on belts, spindles, slides, and coolant pumps
- PC6.** record maintenance activities in machine logbooks as per SOP
- PC7.** inspect cutting tools and inserts for wear, chipping, or deformation
- PC8.** replace or re-sharpen tools as per tool life standards for machining plastics
- PC9.** clean and store tools, tool holders, and fixtures in designated storage to prevent damage
- PC10.** check for abnormal vibrations, sounds, or heating during machine operation
- PC11.** verify coolant/lubricant levels, leakage, and flow during machining
- PC12.** identify and report machine faults, tool breakages, or alignment issues to the supervisor
- PC13.** set tools and accessories in proper working order before production runs
- PC14.** maintain a record of tool usage, replacements, and calibration status
- PC15.** maintain a record of tool usage, replacements, and calibration status

Maintain Work Area and Safety Compliance

To be competent, the user/individual on the job must be able to:

- PC16.** keep the work area free from plastic chips, dust, oil spills, and scrap materials
- PC17.** organize tools, gauges, and accessories in designated places for easy access
- PC18.** ensure gangways, walkways, and machine surroundings are clear of obstructions
- PC19.** segregate usable tools, fixtures, and consumables from damaged or excess items
- PC20.** maintain labeling, tagging, and systematic arrangement of materials as per 5S standards
- PC21.** carry out daily housekeeping checks and report deviations

Qualification Pack

- PC22.** collect, segregate, and dispose of plastic waste, burrs, and scrap as per organizational and environmental guidelines
- PC23.** avoid contamination of reusable raw materials during handling
- PC24.** use safe methods and tools for chip/waste removal to prevent injury
- PC25.** adhere to organizational safety rules, signage, and hazard control measures
- PC26.** follow environmental guidelines for recycling, waste disposal, and storage of materials
- PC27.** respond promptly to safety hazards such as spills, leaks, or fire risks
- PC28.** select and wear appropriate PPE such as gloves, safety glasses, and ear protection during operations
- PC29.** follow lock-out/tag-out procedures during machine cleaning, maintenance, or tool changes
- PC30.** maintain a safe working posture and use ergonomic practices while handling plastic workpieces or machine parts

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the manufacturers guidelines, SOPs, and maintenance schedules for CNC turning machines
- KU2.** methods of cleaning and lubricating CNC machine parts without causing damage
- KU3.** types of cutting tools, inserts, tool holders, and their maintenance requirements for machining plastic
- KU4.** indicators of tool wear, chipping, or failure and their impact on machining quality
- KU5.** importance of monitoring coolant/lubricant levels and their application in plastic machining
- KU6.** common machine abnormalities such as vibrations, overheating, or misalignment and reporting procedures
- KU7.** preventive maintenance practices and how they reduce downtime and extend machine life
- KU8.** workplace 5S practices and their role in efficiency and safety
- KU9.** proper methods of handling, segregating, and disposing of plastic waste and scrap
- KU10.** environmental regulations related to recycling and disposal of plastic materials
- KU11.** organizational safety procedures, signage, and hazard control measures
- KU12.** use, care, and limitations of PPE such as gloves, goggles, masks, and safety shoes
- KU13.** lock-out/tag-out procedures for safe machine isolation during maintenance
- KU14.** ergonomic practices and safe material handling techniques to prevent strain or injury
- KU15.** documentation requirements for maintenance, tool usage, inspection logs, and housekeeping checklists

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** apply basic communication skills to report machine faults, tool wear, or safety hazards to supervisors
- GS2.** follow written instructions, SOPs, and maintenance schedules accurately

Qualification Pack

- GS3.** use digital interfaces, machine HMIs, or logbooks for recording maintenance and inspection data
- GS4.** read and interpret safety symbols, signage, and technical instructions related to machine upkeep
- GS5.** identify common machine issues such as unusual noise, vibration, or overheating and take corrective actions
- GS6.** handle cutting tools, tool holders, and fixtures carefully to prevent damage or injury
- GS7.** apply safe waste handling practices, segregating recyclable and non-recyclable plastic scrap
- GS8.** use appropriate hand tools and cleaning equipment for machine and tool maintenance
- GS9.** work systematically following 5S principles to keep the work area organized and efficient
- GS10.** select and wear appropriate PPE for machine maintenance and housekeeping tasks
- GS11.** practice basic problem-solving when dealing with minor machine, tooling, or work area issues
- GS12.** apply time management skills to complete cleaning, inspection, and tool-setting tasks within allotted time
- GS13.** collaborate with team members to maintain workplace safety and cleanliness
- GS14.** adapt to organizational safety rules and environmental standards consistently
- GS15.** maintain focus, discipline, and responsibility while ensuring machine readiness and work area safety

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain CNC Turning Machine and Tools</i>	20	30	-	-
PC1. remove plastic chips, dust, and debris from machine surfaces and enclosures	1	2	-	-
PC2. clean tool posts, work-holding fixtures, and guards to maintain accuracy and safety	2	2	-	-
PC3. dispose of plastic waste and scrap in designated bins following organizational procedure	1	2	-	-
PC4. check and top up lubrication oil, coolant, or compressed air systems as specified by the manufacturer	1	2	-	-
PC5. Perform routine preventive checks on belts, spindles, slides, and coolant pumps	2	2	-	-
PC6. record maintenance activities in machine logbooks as per SOP	1	2	-	-
PC7. inspect cutting tools and inserts for wear, chipping, or deformation	1	2	-	-
PC8. replace or re-sharpen tools as per tool life standards for machining plastics	2	2	-	-
PC9. clean and store tools, tool holders, and fixtures in designated storage to prevent damage	1	2	-	-
PC10. check for abnormal vibrations, sounds, or heating during machine operation	1	2	-	-
PC11. verify coolant/lubricant levels, leakage, and flow during machining	1	2	-	-
PC12. identify and report machine faults, tool breakages, or alignment issues to the supervisor	2	2	-	-
PC13. set tools and accessories in proper working order before production runs	1	2	-	-
PC14. maintain a record of tool usage, replacements, and calibration status	2	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. maintain a record of tool usage, replacements, and calibration status	1	2	-	-
<i>Maintain Work Area and Safety Compliance</i>	20	30	-	-
PC16. keep the work area free from plastic chips, dust, oil spills, and scrap materials	1	2	-	-
PC17. organize tools, gauges, and accessories in designated places for easy access	2	2	-	-
PC18. ensure gangways, walkways, and machine surroundings are clear of obstructions	2	2	-	-
PC19. segregate usable tools, fixtures, and consumables from damaged or excess items	1	2	-	-
PC20. maintain labeling, tagging, and systematic arrangement of materials as per 5S standards	1	2	-	-
PC21. carry out daily housekeeping checks and report deviations	2	2	-	-
PC22. collect, segregate, and dispose of plastic waste, burrs, and scrap as per organizational and environmental guidelines	1	2	-	-
PC23. avoid contamination of reusable raw materials during handling	1	2	-	-
PC24. use safe methods and tools for chip/waste removal to prevent injury	1	2	-	-
PC25. adhere to organizational safety rules, signage, and hazard control measures	1	2	-	-
PC26. follow environmental guidelines for recycling, waste disposal, and storage of materials	1	2	-	-
PC27. respond promptly to safety hazards such as spills, leaks, or fire risks	2	2	-	-
PC28. select and wear appropriate PPE such as gloves, safety glasses, and ear protection during operations	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. follow lock-out/tag-out procedures during machine cleaning, maintenance, or tool changes	2	2	-	-
PC30. maintain a safe working posture and use ergonomic practices while handling plastic workpieces or machine parts	1	2	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N9906
NOS Name	Maintain CNC Turning machine, tools and work area
Sector	Rubber
Sub-Sector	
Occupation	CNC Machining and Die Manufacturing
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

RSC/N5007: Follow health & safety protocol

Description

This unit is about maintaining health and safety of self and others at workplace.

Scope

The scope covers the following :

- Maintain a clean and efficient workplace
- Render appropriate emergency procedures
- Maintain standard safety procedures at the workplace
- Participate in safety awareness campaigns
- Understand potential sources of accidents
- Use safety gears to avoid accidents

Elements and Performance Criteria

Maintain a clean and efficient workplace

To be competent, the user/individual on the job must be able to:

- PC1.** undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor
- PC2.** work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy
- PC3.** read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc
- PC4.** prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.
- PC5.** use equipment and materials safely and correctly and return the same to designated storage when not in use
- PC6.** dispose off waste safely and correctly in a designated area
- PC7.** risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace
- PC8.** perform work in a manner which minimizes environmental damage
- PC9.** all procedures and work instructions for controlling risk are followed closely
- PC10.** report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger

Render appropriate emergency procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.
- PC12.** follow emergency procedures as per company standards and workplace requirements

Qualification Pack

- PC13.** use emergency equipment in accordance with manufacturers' specifications and workplace requirements
- PC14.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.
- PC15.** recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate
- PC16.** dispose off medical waste in accordance with workplace requirements
- PC17.** report details of first aid administered in accordance with work place procedures

Maintain standard safety procedures at the workplace

To be competent, the user/individual on the job must be able to:

- PC18.** comply with general safety procedures
- PC19.** follow standard safety procedures while handling equipment, hazardous material or tool
- PC20.** check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.
- PC21.** ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure
- PC22.** keep the workplace organized, swept, clean and hazard free

Participate in safety awareness campaigns

To be competent, the user/individual on the job must be able to:

- PC23.** attend fire drills and other safety related workshops organized at the workplace
- PC24.** be aware of first aid, evacuation and emergency procedures
- PC25.** be alert of any events and do not be negligent to any safety procedures to be followed

Understand potential sources of accidents

To be competent, the user/individual on the job must be able to:

- PC26.** avoid accidents while using hazardous chemicals, machines, sharp tools and equipment

Use safety gears to avoid accidents

To be competent, the user/individual on the job must be able to:

- PC27.** use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)
- PC28.** handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies on incentives, delivery standards, and personnel management.
- KU2.** occupational safety and health policy followed
- KU3.** emergency evacuation procedure
- KU4.** medical policy
- KU5.** company laws and acts
- KU6.** the risks to health and safety and the measures to be taken to control those risks in the area of work

Qualification Pack

- KU7.** workplace procedures and requirements for the handling of workplace injuries/illnesses.
- KU8.** basic emergency first aid procedure
- KU9.** local emergency services
- KU10.** reporting on accidents, incidents and problems to appropriate authorities.
- KU11.** how to use machines as per standard operating procedure
- KU12.** how to maintain work area safe and secure
- KU13.** use of hazardous materials, tools and equipments
- KU14.** emergency evacuation and first aid procedures to be followed
- KU15.** personal hygiene and fitness requirements
- KU16.** general duties under the relevant health and safety legislation
- KU17.** what personal protective equipment and clothing should be worn and how it is cared for
- KU18.** the correct and safe way to use materials and equipment required for work
- KU19.** the importance of good housekeeping in the workplace
- KU20.** safe disposal methods for waste
- KU21.** methods for minimizing environmental damage during work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data which are required for record keeping purpose
- GS2.** report problems to the appropriate person in a timely manner
- GS3.** write descriptions and details about incidents in reports
- GS4.** read instruction manuals for hand tools and equipment
- GS5.** read instructions on work orders and procedures
- GS6.** receive instructions and seek advice from superiors
- GS7.** communicate clearly and effectively with others
- GS8.** take a decision for any change/issue based on earlier successes(documented previous history)on similar issues
- GS9.** work out changes in case a new improved machine/equipment is added in the process or any new material/chemical is developed replacing existing one.
- GS10.** make changes in cycle time due to improved process.
- GS11.** use the standard operating procedure or trouble shooting manuals for trouble shooting and other reference documents approved by plant management
- GS12.** consult the peer group and superiors to arrive at a favourable decision.
- GS13.** use of standard available problem solving techniques for decision making
- GS14.** review and analyze the process steps to check on system non adherence andnon conformity
- GS15.** review the current sop and other standards for continuous improvement to facilitate decision making
- GS16.** take a calculated risk with minimum losses
- GS17.** schedule daily activities and drawing up priorities; allocate start times, estimation of completion times and materials, equipment and assistance required for completion.

Qualification Pack

- GS18.** match customer needs/specification by adjusting the processing conditions (interact with customer in case any clarification required)
- GS19.** ensure that performance of his action/operation/activity does not lead to any divergence from the specified quality of the final product as required by the customer.
- GS20.** complete the assigned task in timely manner so that the final product is delivered in the timeline given by the customer.
- GS21.** communicate effectively to the superior/customer for any delay in supplies to the clients.
- GS22.** work towards fulfilling the customers requirement as per their demand.
- GS23.** in case of any complaint, ensure its timely resolution if the problem is emanating at his level
- GS24.** communicate effectively to the superior/customer for any delay in resolving the problem faced by the customer.
- GS25.** maintain good/cordial relation with customers.
- GS26.** work on the feedback received from customer regarding the product.
- GS27.** use first aid treatment in case of any injury/accident.
- GS28.** monitor and maintain the condition of tools and equipment
- GS29.** assess situation & identify appropriate control measures
- GS30.** act, communicate and report in emergency situation

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain a clean and efficient workplace</i>	9	21	-	-
PC1. undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor	2	2	-	-
PC2. work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy	2	2	-	-
PC3. read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc	-	2	-	-
PC4. prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.	2	2	-	-
PC5. use equipment and materials safely and correctly and return the same to designated storage when not in use	1	2	-	-
PC6. dispose off waste safely and correctly in a designated area	2	4	-	-
PC7. risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace	-	2	-	-
PC8. perform work in a manner which minimizes environmental damage	-	2	-	-
PC9. all procedures and work instructions for controlling risk are followed closely	-	1	-	-
PC10. report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger	-	2	-	-
<i>Render appropriate emergency procedures</i>	9	18	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.	2	4	-	-
PC12. follow emergency procedures as per company standards and workplace requirements	2	4	-	-
PC13. use emergency equipment in accordance with manufacturers' specifications and workplace requirements	2	4	-	-
PC14. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.	-	1	-	-
PC15. recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate	-	2	-	-
PC16. dispose off medical waste in accordance with workplace requirements	-	1	-	-
PC17. report details of first aid administered in accordance with work place procedures	3	2	-	-
<i>Maintain standard safety procedures at the workplace</i>	6	15	-	-
PC18. comply with general safety procedures	2	4	-	-
PC19. follow standard safety procedures while handling equipment, hazardous material or tool	-	2	-	-
PC20. check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.	2	4	-	-
PC21. ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure	-	1	-	-
PC22. keep the workplace organized, swept, clean and hazard free	2	4	-	-
<i>Participate in safety awareness campaigns</i>	2	8	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. attend fire drills and other safety related workshops organized at the workplace	1	3	-	-
PC24. be aware of first aid, evacuation and emergency procedures	1	3	-	-
PC25. be alert of any events and do not be negligent to any safety procedures to be followed	-	2	-	-
<i>Understand potential sources of accidents</i>	1	3	-	-
PC26. avoid accidents while using hazardous chemicals, machines, sharp tools and equipment	1	3	-	-
<i>Use safety gears to avoid accidents</i>	3	5	-	-
PC27. use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)	2	2	-	-
PC28. handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders	1	3	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N5007
NOS Name	Follow health & safety protocol
Sector	Rubber
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

Qualification Pack

6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass 50% aggregate for the QP.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
RSC/N9904.Interpret CNC Programs and drawings and set up CNC Turning Machine	40	60	0	0	100	25
RSC/N9905.Perform turning operations on Plastic Workpieces using CNC Machine	40	60	0	0	100	25
RSC/N9906.Maintain CNC Turning machine, tools and work area	40	60	0	0	100	25
RSC/N5007.Follow health & safety protocol	30	70	-	-	100	15
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	170	280	-	-	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.